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Housing ownership Gender differences in Dhaka, Bangladesh☆

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1. Introduction

Property ownership and rights frame individuals' economic behaviors related to investment and household purchase, which, in turn, affect broader economic development. Although property is widely recognized as an important resource for empowering women and many development policies worldwide call for strengthening women's rights to property, little is known about the gender differences in housing ownership, particularly in urban areas. Housing is critical to people's identity, dignity, and sense of belonging and for moving out of poverty (Chant 2013) and is a key asset for women as a store of wealth and means of generating income (World Bank et al., 2015). Galiani and Schargrodsky (2011) suggested that property rights can affect the efficiency of resource allocation. Land titles enhance the incentives to invest by reducing the risk of the land being taken away and by allowing resources that would be otherwise used to protect the property itself to be redirected. Moreover, titles may also enhance credit transactions through trade in assets and collateralization. Property rights, therefore, should be seen in terms of the extent to which women enjoy equal rights with men, and addressing any gap should be a priority of tenure policy (UN-Habitat (United Nations Human Settlements Programme), 2004). Policy makers and researchers agree that strengthening women's property rights is essential for poverty reduction and equitable growth (Kieran et al. 2015).

Recent research on gender and housing reveals that, in many developing countries, women and men access land and housing differently (Klugman et al. 2014; Kieran et al. 2015; O'Sullivan 2017; Gaddis et al. 2018). Women lag men in the opportunity to acquire or hold housing assets because of customary and religious laws, traditional inheritance practices, and marital property regimes. Data from 13 developing countries suggests that women are less likely to report owning land or housing than men (Klugman et al. 2014).

What's more, asset ownership data is often available at the household-level. Even when data is disaggregated at the individual level, respondents of surveys are usually proxies, often the male household head, who may not report accurate information, leading to inferior data quality. In this regard, individual-level ownership data could provide insights into women's economic empowerment (Doss et al., 2017, Kilic and Moylan., 2016, Kieran et al. 2015, Kabeer 2013).

Bangladesh is a populous developing country that has made great strides in poverty reduction, and, at the same time, has seen greater gender equality in many aspects, such as reduced fertility rates and achieved gender parity in schooling. According to the World Bank's Human Capital Index, girls in Bangladesh have a better chance than boys to complete school and survive to age 60. Women's labor force participation rates have increased from 26 percent in 2003 to 36 percent in 2016. However, large gender gaps remain. Women still lag behind men in terms of employment, access to finance and asset ownership (Farole and Cho 2017; Solotaroff et al. 2019). At the same time, the country is urbanizing fast; in 2018, its urban population was 37 percent of the total population and is projected to reach 58 percent by 2050 (UN DESA 2018). Meanwhile, the absolute number of extreme poor has been increasing in urban Bangladesh (Raza 2018). The Bangladesh Integrated Household Survey, conducted in rural areas by the International Food Policy Research Institute in 2015, showed a large gender gap in housing ownership in rural areas (Solotaroff et al. 2019). However, data about housing ownership in urban areas are lacking. Moreover, while evidence from other countries also suggest that risk of eviction could affect women and men differently (Field 2005, Desmond 2012, O'Sullivan 2017), little data is available in Bangladesh.

Our paper uniquely contributes to the data gap about gender equality in housing ownership in urban areas. In 2018, we conducted the Dhaka Low Income Area Gender, Inclusion, and Poverty (DIGNITY) survey. We collected individual-level data on men and women's housing ownership in low-income areas of Dhaka, the capital of Bangladesh. We randomly selected about 1,300 households and employed an innovative survey

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methodology: two respondents—one male and one female with the most knowledge of household matters (mostly the primary couple of the household)—were interviewed separately by interviewers of the same sex. The survey instrument asked about employment, migration history, and empowerment, along with housing ownership. With some modifications to fit the urban setting of Bangladesh, our survey instrument and methodology followed recommendations by Kilic and Moylan (2016), distinguishing between reported owners, economic owners, and documented owners and holders of various bundles of rights. Individuals might not be legal owners of property, but still enjoy decision-making power over how the property is used or to whom it is bequeathed (Gaddis et al. 2018). The property rights collected were the right to sell, bequeath, rent, and use the property as collateral for borrowing. These rights provide value added because broad definitions of ownership–reported, documented, and rights-based–would help stakeholders understand how asset ownership matters for the men and women.

Our data analysis reveals interesting gender patterns: a relatively high level of gender equality in the rates of reported housing ownership. About 9.7 percent of male and 9.2 percent of female respondents reported that they are among the owners of their dwelling. The reported ownership rates are also in line with other dimensions of rights (such as rights to sell, rent out, and bequeath), which are quite similar between men and women. Given the global patterns of large gender gaps in property and housing ownership found in rural Bangladesh and in many countries (Klugman et al. 2014; Kieran et al. 2015; O'Sullivan 2017; Gaddis et al. 2018, Solotaroff et al. 2019), the figures from urban Bangladesh underscore the need to better understand the gender dynamic in the urban poor populations as well as implications for development outcomes.

We further examined how men and women report their risks of eviction and found that women express greater concern about eviction than men (22 percent vs. 15 percent). We also found that documented ownership[1] is correlated with perceived tenure security: the reduction in concerns about eviction among women with documented ownership is more than double that of men. For women, having any forms of housing ownership document is associated with a 23 percentage points reduction in the concern about eviction, compared to an 11 percentage points reduction for men. We think this finding has crucial policy implications.

In the following sections, we present a review of literature on topics related to gender equality in housing ownership (section 2). Then we describe the data collected for this research (section 3); present descriptive findings, such as the level of gender differences in housing ownership and rights; (section 4); and, finally, discuss conclusion and the policy implications (section 5).

2. Literature Review

This section presents the literature related to gender equality in property ownership in developing countries, starting from literature about ownership and property rights and the impacts of gender equality in housing ownership on development outcomes, followed by the extent of gender gaps across countries and the context of urban housing in Bangladesh.

2.1. Gender gaps in housing and property ownership around the world

Ostrom and Hess. (2008 11) define a property right as "an enforceable authority to undertake particular actions in specific domains. The rights of access, withdrawal, management, exclusion, and alienation can be separately assigned to different individuals as well as viewed as a cumulative scale moving from the minimal right of access through possessing full ownership rights." Property rights guide behaviors among individuals. If one individual has a right, someone else has a duty to observe that right. In this regard, "Owners" possess the right of alienation—the right to transfer a good in any way the owner wishes that does not harm the physical attributes or uses of other owners—in addition to the bundle of rights held by a proprietor (Schlager and Ostrom., 1992). Understanding the definition of ownership in each setting requires an understanding of the rights associated with it. Although documented ownership is often considered the most secure, it only applies to certain assets and in places where enforcement of the associated rights is effective (Doss et al. 2017). The rights also underscore the role of property in generating income. However, some people may only have some of the rights in the bundle (Doss et al. 2015).

Gender equality in ownership of productive assets, such as land and housing, is important to economic development and women's empowerment. Agarwal (1994) notes that the most important economic factor affecting women in South Asia is the gender gap in command over property. Gender gaps in property ownership generate inefficient allocation of assets. Insecure property rights may cause women to underinvest in their land and thus receive lower agricultural yields than their male counterparts (O'Sullivan 2017). Higher levels of property ownership among women in urban compared to rural areas are usually attributed to greater social and economic opportunities (Chant 2012). In Peru, joint titling allows urban women to seek paid work, instead of spending time safeguarding their land against property invasion (Field 2007). For many poor urban women in India, a house is a means of income generation: As a large proportion of them earn income either at home or near their homes, housing ownership with basic services can increase female labor force participation (Baruah 2007). A long-term longitudinal study from rural Tanzania reveals that women's property and inheritance rights are significantly correlated with their employment outside the home, self-employment, and earnings (Peterman 2011). A study in Bangladesh also shows that ownership of residential land and housing are positively correlated with empowerment indicators, such as decision making in the household, mobility, and participation in local politics (Kabeer 2013).

In terms of access to finance, a cross-country study revealed that countries whose marital regime is community property [2] tend to have higher proportions of females with financial accounts (Iqbal 2015). A study of women's property rights in regularized slums in Chandigarh, India showed that 30 percent of women are more actively involved in making decisions about finance, while as many as 65 percent felt empowered because of joint legal titles with their husbands that prevent the unilateral sale of the house. The study also found that a joint title increases the probability of a woman asserting herself by 13 percent and being in a regularized settlement with services and tenure security increase the likelihood that a woman would report that she would protest the sale of her house by 20 percent (Datta 2006). In Ethiopia, joint titling increases women's participation in land rental markets, possibly because of a lower perceived risk of expropriation, as well as enhances knowledge of land rights and local political participation (O'Sullivan 2017). In Karnataka state of India, women's ownership of a house or land improves women's autonomy in decision making but does not impact egalitarianism in decision making between couples (Swaminathan et al., 2012a, 2012b). Ownership of a house or a plot of land enhances women's ability to travel to the market, health center, and other places outside the community, and to make decisions about their employment, health, and use of money independently. (Swaminathan

[1] It should be noted that the ownership definition in this analysis is the perceived ownership based on documents that can be used as evidence of occupancy rights; the rights to rent out, sell and bequeath underline perception of ownership.

[2] Among marital property regimes, the community property regime, in which both spouses equally share property, is considered more gender equal than other regimes such as the separate property regime, in which each spouse retains ownership of their own property.

et al., 2012a, Swaminathan et al., 2012b2012b). In Nepal, female landowners have more power over household decisions and their children are better nourished (Allendorf 2007). In Peru and Argentina, urban squatters who received titles tend to experience lower fertility rates (Galiani and Schargrodsky., 2011). Finally, property ownership also reduces vulnerability for women against risk factors such as male migration, abandonment, divorce, polygamy, death, or natural disaster, particularly when women acquire property rights through relationship with men. Women's property ownership is even more important in urban settings, where they may have less access to support from their family of origin (Rabenhorst 2011).

Women face greater barriers in owning housing than men. Many tenure regimes— including statutory, customary, and informal—discriminate against women either legally or in practice. In some cultural traditions, women are expected to depend on their fathers, husbands, or sons for tenure security, denying the women's basic rights, especially for those who are divorced or widowed (UN-Habitat (United Nations Human Settlements Programme), 2004). Women tend to have limited income capability and access to finance compared to men; thus, they are less likely to be able to afford housing. Women also face gender-based discrimination and violence. Legislation—particularly family law, inheritance law, and land law—may curb women's rights to own and control property. Even where laws are equal for both men and women, customary and religious practices interact and govern women's property rights. Social norms could further restrict women's land ownership and their ability to exercise agency over land and housing. The practice of patrilocalism and social norms that dictate men as the sole decision makers in the household may prevent gender-equal laws from being implemented (Alorro and Kotikula., 2016).

Globally, existing data show large gender property ownership gaps favoring men. In this regard, existing literature mostly discusses land ownership due to greater data availability (Kieran et al. 2015, O'Sullivan 2017, Gaddis et al. 2018), with limited information about gender gaps in housing ownership (Doss et al. 2011, Gaddis et al. 2018). In low-income urban communities in Ghana, Senegal, Tanzania, Uganda, Zambia, Sri Lanka, Colombia, and Costa Rica, a study by Miraftab (2001) found that although about 62 percent of all households in all the communities are homeowners, only about 30 percent of those owners are women. In Sub-Saharan Africa, Gaddis et al. (2018) analyze DHS data from 28 countries and find a large gender gap for housing, with 43 percent of women claiming sole or joint ownership compared with 55 percent of men. They also observe that there is little difference in the gender gaps between land and housing. The Gender Asset Gap project shows how patterns of gender gaps in housing ownership differ across three countries. In Ghana, the rate of female housing ownership is 16 percent compared to 30 percent among men. The ownership rates of men and women are quite equal in Ecuador, with a female ownership rate of 36 percent and a rate of 34 percent among men. (About a quarter of households in Ecuador are headed by a nonpartnered woman, which could explain the relatively high rate of individual dwelling ownership by women.) However, the gap is much larger in the Indian state of Karnataka: only 16 percent of dwellings are owned jointly or solely by women, compared to 42 percent for men (Doss et al. 2011).

The Bangladesh Integrated Household Survey, conducted in rural areas by the International Food Policy Research Institute in 2015, revealed that 93 percent of rural men reported ownership of dwelling, compared to only 18 percent for rural women (Solotaroff et al. 2019). But data about housing ownership in urban areas are still limited, which is unfortunate given the high urbanization rate of the country. We may expect different gender patterns when comparing rural and urban areas. In rural areas, land and housing are often considered as a combined asset (Gaddis et al. 2018). It may be harder for women to own land in heavily agrarian economies, where land is increasingly fragmented into more inheritances due to its high fertility; thus, ownership may be contested. But in urban areas, particularly in informal settlements, housing is less likely to be acquired through inheritance; so there is greater opportunity for women to own. We therefore expect that the gender gap in housing ownership in urban areas will be smaller than what has been observed in rural areas and even smaller than the gender gap in land ownership.

Among the urban poor, the risk of forcible eviction amplifies poverty as it discourages investment and housing improvement (Nakamura 2017). Evidence from the United States suggest that women from inner-city neighborhoods are evicted at significantly higher rates than men (Desmond 2012). Fear of eviction in urban informal settlements could cause households to underinvest in housing improvement, which is particularly costly for urban neighborhoods threatened by natural disasters such as floods (Field 2005). Women's insecure property rights may also cause them to underinvest (O'Sullivan 2017). However, there is still a gap in the literature about differential gender impacts on risks of eviction and our review did not find existing literature about how gender equality in housing ownership may affect perception of eviction risks.

2.2. *Bangladesh context: Urbanization, tenure security and inheritance*

2.2.1. *Urbanization*

Bangladesh is urbanizing fast. In 2018, its urban population was 37 percent of the total population and is projected to reach 58 percent by 2050 (UN DESA 2018). The trend suggests that most of the population increase will occur in urban areas. Concurrently, the slum population has increased around 12 percent per year, nearly double the population growth in urban areas (BBS 2014). Many women are actively taking part in this urbanization process through rural-urban migration, enabling them to participate in the cities' labor force (Solotaroff et al. 2019).

Dhaka, the capital of Bangladesh, is inhabited by 18.6 million people and was the most densely populated city in the world in 2019 (Demographia 2019). Within the city corporation limit, there are 3,399 slums (BBS 2015). The Bangladesh Bureau of Statistics (BBS) defines a "slum" as a cluster of compact settlements of five or more households that unsystematically and haphazardly expand in an unhealthy condition and atmosphere on government and private vacant land. A slum generally includes small housing units built with poor and temporary materials, high population and structure density, and lack of access to basic services such as sanitation and drinking water.

2.2.2. *Tenure Security*

A typical slum in Dhaka tends to be formed through squatting on public land. Most households in such communities derive their livelihoods from both informal and formal sector employment. Work includes rickshaw pulling, small-scale trading, domestic service, garment factory labor, and administrative employment in the public or private sector. Over the years, slums tend to have evolved into self-contained neighborhoods, with their own facilities such as markets, primary and secondary schools, and so on. Services such as electricity and water are often obtained through illegal connections. Although they do not have formal land rights, many residents built their houses when the land was first occupied. They are considered "owners" of their dwellings, and most now own more than one dwelling while the rest rent from these original squatter "owners." Still, these landlords are vulnerable to eviction. To defend themselves from government development plans, residents organize into community groups, combining local informal civil action and patronage (Lewis and Hossain., 2017).

Female landlords, as discussed in Lewis and Hussain (2017), are active in community-based organizations, which are often set up by nongovernment organizations (NGOs). These groups motivate the residents to demand improved provisions and access to services. Landlords are typically quite proactive when engaging with NGOs, as the relationship aids their goals to improve not only their own households, but also those of renters, in hopes of increasing the rental value.

Land administration in Bangladesh is relatively complex, involving three different ministries: the Ministry of Land; the Ministry of Law, Justice and Parliamentary Affairs; and the Ministry of Public

Administration. Updating record-of-rights and land transfer registrations are undertaken by administrative units within these ministries. Such a complex system produces a number of challenges, including a lack of coordination between ministries and agencies. Because the process is not electronic, there are large backlogs and delays, as well as the potential for errors due to both fraud and unintentional errors. Cumbersome processes create the opportunity for officials to demand bribes. Corruption within administrative units increases costs for citizens and prevents the poor and traditionally marginalized groups, like women, from accessing land titles. For example, land administration alone accounts for almost 40 percent of the total bribe paid at the local level when property owners either try to transfer titles or address any other issue related to their land parcels (Rahman 2014).

2.2.3. Inheritance

Inheritance rights affect gender equality in Bangladesh and neighboring countries. Land is mainly inherited in Bangladesh (as opposed to being acquired through the market or by government transfer). For example, 78 percent of rural households acquire agricultural land through their parents (Solotaroff et al. 2019).Women's property rights and inheritance in Bangladesh are highly controversial (Rahman 2014), and gender gaps in inheritance vary by location and economic class. Solotaroff et al. (2019) indicates that rural women mentioned inheriting land disproportionately less often than urban women.

In addition to custom and norms, parents' lack of awareness of their daughter's capability to manage property for economic gain may prevent them from bequeathing land to their daughters. Economic activities are seen as a man's role only, and many rural families believe that female heirs would not be able to utilize their property if they inherited it (Solotaroff et al. 2019). Moreover, because forfeiting their inheritance is traditionally considered to be the proper and respectable thing to do, women in Bangladesh give up their shares of inheritance to brothers, even though it has been already allocated for them. This pressure is tied to the practice called *Naior,* which allows women to visit their natal family and ensures them future protection in case of conflict in the marital home (Sourav 2015).

Religion, on the other hand, is an enabling factor for women's inheritance. According to Islamic law, women are entitled to half of their brothers' inheritance. Solotaroff et al. (2019) suggests that religious fathers tend to insist on bequeathing some land to their daughters. A similar pattern is also found in Pakistan (Ahmad et al., 2012), especially if the fathers learned about the law from formal religious sources or they had formal religious education. However, despite Islamic law, which applies to about 90 percent of the Bangladeshi population, women still inherit far less property (Solotaroff et al. 2019).

3. Data and method

The Dhaka Low Income Area Gender, Inclusion, and Poverty (DIGNITY) survey was designed to shed light on poverty, economic empowerment, and livelihoods in Bangladesh's urban areas. The array of information collected ranged from demographic and socioeconomic characteristics of men and women in households, non-cognitive skills, time use, attitudes, perceptions of work, and detailed work history. The DIGNITY survey is representative of low-income areas of the Dhaka City Corporation (CC) and the Greater Dhaka Statistical Metropolitan Area, following a two-stage stratification design using the probability proportional to size methods. The survey covered 18 slum and 47 nonslum primary sample units (PSUs). The sampling frame of the nonslum group was based on the Dhaka CC population census of 2011, while the slum group was based on Dhaka CC population census from 2013–14. In total, the sample was composed of 1,300 households and 2,378 individuals, with 1,713 from nonslum and 665 from slum areas.

Traditional surveys only interview the head of households (who tend to be men) and ignore information from the female household members. The key feature of the DIGNITY survey is its direct data collection from household members. The survey instrument is composed of two main modules: a traditional household module (in which the household head is interviewed on basic information about the household) and an individual module, which interviews respondents individually. Two persons from each household—one male and one female—mostly the main couple, were selected for the interview. Consequently, the survey team deployed a male and a female interviewer per each household, so that the interviewer's gender matched that of the respondent.

Our methodology for collecting data about ownership and rights of land and housing in the DIGNITY survey responds to earlier calls for collecting data on ownership of and rights to assets at the individual level (Grown and Gupta, 2005; Doss et al. 2011) and follows recommendations in Kilic and Moylan (2016). Their recommendations include "1) reducing the reliance on a single respondent; 2) expanding the practice of interviewing multiple age-eligible individuals per household, with a focus on the members of the principal couple if a couple is present; and 3) probing directly regarding respondents' personal ownership of and rights to assets, whether exclusively or jointly, with someone else" (Kilic and Moylan., 2016, 23).

Individual data collection with direct interviews of female subjects could improve data accuracy for personal ownership of assets, income, and housing value because it captures women's ownership better than a proxy interview of the most knowledgeable person in the household (most likely the head). Proxy respondents may not provide adequate information on the income and labor inputs of others, including that of their spouse (Doss et al., 2017).

Data collection on wealth must consider the sex of the respondent (Doss et al. 2018). The fact that women may be disadvantaged in both statutory and customary land-tenure systems further supports the argument for collecting data about property ownership and rights at the individual level (Kieran et al. 2015). Data from the Gender Asset Gap Project in Ecuador, Ghana, and the Karnataka state of India suggest that men and women respond differently to questions about housing values: The distributions of values reported by men are wider than those reported by women. In Southern Malawi, Fisher et al. (2010) found that interviewing only the husband can lead to underestimating wife's income and does not accurately estimate the total income in most households. In Uganda, Kilic and Moylan (2016) found that when the survey asked directly about personal ownership of assets, female respondents were more likely to identify themselves as owners of the dwelling, livestock, and financial assets. For example, directly interviewing respondents rather than only the proxy increases on average the incidence of joint reported dwelling ownership by 11.4 percentage points (or the marginal effect corresponds to 81 percent).

The main variables of interest in our paper are ownership and rights over housing. The data were collected by interviewing each respondent separately about dwelling ownership (reported or perceived), ownership documents, and the bundle of rights, including the right to sell, to bequeath, to use as collateral, to rent out, and to make improvements. The survey also collected information about documented ownership by asking if the respondents had an ownership document for the dwelling, which could be formal (such as a title deed) or informal (such as a utility or tax bill). But because of the informal settlement, a "formal" document is not necessarily an official document that grants all the rights associated with one. Thus, for our analysis, we used a single variable for documented ownership when the individual has any type of document. This ownership definition own is essentially the perceived ownership based on documents that can be used as evidence of occupancy rights.

Another outcome variable is the perceived tenure security, which is an important topic in urban housing. Housing units are generally constructed on land to which the occupants have a legal claim due to investment costs and risk avoidance. However, the housing situation for low-income residents in developing countries may veer off that norm

and fall into the realm of informal housing, or housing built on land to which the occupants do not have legal claim.[3] Informal housing may be rented out for much less than formal rental units, which makes it appealing for low-income individuals and families. However, informal urban housing may face risks of eviction to varying degrees that range from pavement dwellers, who face the highest risk of eviction, to the "grey zone," which may be secure enough for short-term access to shelter and services, but over the long-term face risks of eviction, such as in the cases of squatters, residents of unauthorized subdivisions, and even legal owners of unauthorized construction. The housing tenure types with low eviction risks are tenants with a contract, lease-holders, and free-holders or owners. These tenure types face the lowest risk of eviction (Payne et al., 2014; Barry and Augustinus., 2016). In Bangladesh context, risk of eviction is the risk of expropriation by government and private landowners, with small probability of adequate compensation or relocation. While the eviction is traumatic, the fear of eviction may be even more harmful as the vast majority of slum dwellers do not wish to return to their villages (Mohit 2012). In this regard, the DIGNITY survey asked respondents about the security of their dwelling situation (*are you concerned about being evicted),* to which respondents could answer "yes" or "no." It should be noted that this question captures the perception of security of their living situation and their property (in the case of housing owners), or "perceived" tenure security.[4] Summary statistics for each ownership and rights over housing and other variables of interest are depicted in Table 1. Overall, about 9.5 percent of respondents identified themselves as reported owners of their dwelling, but only about 3.5 percent had some forms of documentation about their housing ownership. Finally, about 19 percent of respondents reported that they were concerned about being evicted.

3.1. Summary statistics of housing and other characteristics

Table 1 shows that, on average, respondents' age was 35 years. More than 90 percent were married. The education level of respondents was about 3.6 years of schooling. On average, about 68 percent of respondents had lived in the community for longer than 5 years.[5]

In terms of household-level characteristics, the household size was about 3.9 people per household, and about half of the members were men. Overall, slum residents make up about 16 percent of the survey population. Housing quality in low-income areas of Dhaka is characterized by limited space but decent access to services. On average, a single household occupies a space of about 122 square feet, and about 77 percent of households have a separate kitchen (which could be communal, located away from the main quarter). Similarly, 88 percent of the households had access to improved toilets, but 90 percent of these were shared with other households. Other amenities include access to improved water sources (95 percent) and access to electricity (100 percent).

Women are relatively active in labor markets. The female labor force participation rate (FLFP) from the DIGNITY survey was 58 percent, while that of males was 97 percent. This rate is much higher than the urban average from the 2016 Labor Force Survey, in which the labor force participation rate for females was 33 percent and for males 94 percent.

Although variables about women's economic empowerment were not used in the econometric analysis, we think it is still illustrative to present in this paper, given the novelty of this dataset. Moreover, the extent of gender inequality will provide context for readers about the status of women in low-income communities. Women, on average, still have limited mobility and spend more of their time at home. About 24 percent of working female respondents work from home, while about 41 percent work within their community.[6] This pattern sharply contrasts with that of men, in which almost half work outside the community. The DIGNITY survey also collected data about time use, using the one-day recall method. Women spent more time at home than men (20 hours per day versus 14 hours per day), while men spent more time outside home but in the community (4.5 hours versus 2.2 hours) and outside the community (5.6 hours versus 1.4 hours per day) (see Table 1). In terms of activities, on average women spent about 4 hours per day on employment-related activities, and men spent about 10 hours per day on these activities.

Table 1
Summary Statistics.

Variables	Mean	Std. Dev.	Obs
Housing: reported ownership	0.09	0.29	2,376
Housing: documented ownership	0.04	0.18	2,278
Housing: right to sell	0.06	0.23	2,376
Housing: right to bequeath	0.05	0.22	2,378
Housing: right to use as collateral	0.05	0.22	2,376
Housing: right to rent out	0.05	0.23	2,372
Housing: right to make improvement	0.05	0.23	2,378
Respondent concerned about eviction (1/0)	0.19	0.39	2,378
Age of respondent	34.95	11.63	2,378
Age at marriage	18.85	4.68	2,378
Married (1/0)	0.91	0.29	2,374
Years of completed education	3.55	3.70	2,378
Father's education (years)	1.79	3.44	2,374
Mother's education (years)	0.91	2.19	2,374
Household size (number of persons)	3.94	1.55	2,378
Proportion of male members in household	0.51	0.17	2,378
Live in a slum (1/0)	0.16	0.37	2,378
Duration in current community: More than 5 year (1/0)	0.68	0.47	2,378
Total available space in hh (sqft)	122.42	49.57	2,378
Housing: Has a separate kitchen (1/0)	0.77	0.42	2,378
Roof material: tin (1/0)	0.71	0.45	1,719
Roof material: brick/cement (1/0)	0.26	0.44	1,719
Housing: Has access to improved toilets (1/0)	0.88	0.33	1,719
Housing: Toilet shared with other households (1/0)	0.90	0.30	2,376
Housing: Gets water from an improved source (1/0)	0.95	0.21	2,376
Housing: Household has electricity (1/0)	1.00	0.05	2,376
Monthly per capita food expenditure (Taka)	832.64	710.08	2,376
Individual is in the labor force (1/0)	0.77	0.42	2,378
Emp location: home (1/0)	0.15	0.36	2,378
Emp location: community (1/0)	0.36	0.48	2,378
Emp location: outside the community (1/0)	0.49	0.50	2,378
Hours spent at home per day	17.22	5.88	2,378
Hours spent outside home but within the community per day	3.31	4.82	2,378
Hours spent outside the community per day	3.48	5.44	2,378

3.2. Research method

This paper aims to answer two research questions: do women in low-income areas of urban Bangladesh own housing in the same way as men and is female housing ownership associated with perceived tenure security? For the first question, we present descriptive findings about gender differences in housing ownership and rights. To better

[3] https://stats.oecd.org/glossary/detail.asp?ID=1351.

[4] Barry et al. (2015) suggests that "perceived" tenure security could be relevant to household wellbeing in some situations. For example, a long-term slum resident without formal documentation as his or her home's proof of purchase do not have formal security, yet they believe—because they are involved in a community organization to fight eviction—that they are highly unlikely to be evicted.

[5] Numbers reported are weighted averages of all male and female respondents.

[6] Compared to the national average, the rate of women working outside home in the DIGNITY sample is high. The 2016 Labor Force Survey showed that among working women, only 38 percent work outside home (Islam and Kotikula, 2019).

understand the gender gaps in housing ownership and rights, we explored factors associated with housing ownership, the right to sell the housing, bequeath it, use it as collateral, rent it, or carry out improvements. In each instance, the linear probability model specification was used. The coefficient of a continuous explanatory variable is the effect of a 1-unit change resulting in the percentage change of the dependent variables, holding all else constant. The coefficients of binary explanatory variables can be interpreted as the percentage point change in the dependent variable when the explanatory variable switches from 0 to 1, holding all else constant.

Factors explored in these models included variables that theoretically are likely to be correlated with ownership of property. To the extent possible, we relied on empirical literature to select factors in the estimation model. Because age of marriage is often used as a proxy for women's bargaining power within the household this could affect ownership status. Older persons will have a longer opportunity to accumulate assets, which is shown to be correlated to asset ownership in Bangladesh (Solotaroff et al. 2019). Education level could affect earning potential and wealth accumulation. In addition, the education level of a respondent's father and mother could signal wealth of respondent's natal family. Employment status has also been shown to be correlated with female asset ownership in many settings (Baruah 2007; Peterman 2011; Chant 2012; Swaminathan et al., 2012a, 2012b). People who lived in the community for a long time were likely to have built their houses when the land was first occupied, which is also applicable to respondents living in slum communities. The total available space of the dwelling was expected to be correlated with ownership status because people who are original owners could be living in a larger dwelling, while renters, who arrive at the community later, may only rent small rooms from landlords (Lewis and Hossain., 2017). Ownership status is expected to be correlated with the monthly per capita food expenditure, which is a measure of monetary wellbeing. Household size and the share of male household members are expected to negatively correlate with the ownership of women as women may face less competition to control resources in smaller households (Solotaroff et al. 2019).

For the second research question, we used the linear probability model specification to test whether housing ownership decreases the perceived risks of eviction among women. The dependent variable was the concern about eviction as reported by respondents (a dummy variable), and the explanatory variables were the housing ownership variables: reported ownership and documented ownership. In order to examine the differential effects of housing ownership between men and women, we estimated each model twice, first with a housing ownership variable and then with the addition of an interaction term between ownership and female dummy variables. As data from the United States suggests that women face greater risk of eviction than men (Desmond 2012), the sex of respondents was included in the model specification in order to examine gender gaps in perceived risks of eviction. Similar to the model for the first research question, control variables were included in the model specification. Slum and location characteristics are included to control for the insecurity in informal settlements. Size of dwelling and length of stay in the community were correlated with participation in community-based organizations, which could help protect property against eviction (Lewis and Hossain., 2017). We also included per capita food expenditure to capture poverty status of the household. Finally, we controlled for individual-level and household-level characteristics; namely, labor force participation, age, education, marital status, age of marriage, household size, and composition. The role of family size reflects the fact that the more individuals are living in a household, the more likely it is that someone stays at home at a given time regardless of security considerations.

4. Results

The DIGNITY data revealed important insights about housing ownership and economic lives of residents in low-income areas of Dhaka, Bangladesh. In this section, we first present patterns of ownership in slum and nonslum communities, as well as the gender patterns in housing ownership. Subsequently, we describe nuances of ownership (reported versus documented) as well as economic rights over dwellings. In the second half of this section, we analyze the correlates of women's housing ownership, using a linear probability regression model to tease out key factors that are associated with this. Finally, we explore whether reported and documented housing ownership are linked to the risk of eviction.

4.1. Gender gaps in housing ownership and rights

To answer the first research question "do women in low-income areas of urban Bangladesh own housing the same way as men," we present the proportions of male and female respondents who are owners of their dwellings. The DIGNITY data allow us to disaggregate reported ownership as well as reported economic rights by sex of respondents. The data showed that the pattern of reported housing ownership (either sole or joint ownership) is relatively equal between sexes. About 9.2 percent of female and 9.7 percent of male respondents reported that they are among the owners of their dwellings (Table 2).

We also compare the gender housing gap from this study with the available data from other countries. Calculating the ratio of female housing ownership to male housing ownership, the gender housing gap ratio from DIGNITY survey is about 95 percent. This is much smaller than the gap found in rural Bangladesh (20 percent) and in the Indian state of Karnataka (38 percent) and also higher than the gap ratio in Africa calculated from Gaddis et al. (2018), which is around 78 percent. However, the Dhaka gap ratio is close to that of Ecuador (106 percent). This could suggest the importance of an urban setting as well as greater economic opportunity for women in sectors like the garment sectors in urban Bangladesh. The rates of reported housing ownership are also in line with other dimensions of housing rights, such as the rights to sell, bequeath, use as collateral, rent out, and make improvements. In this regard, we examined reported property rights of respondents who are reported owners. Table 3 shows equality of such rights between men and women. About 60 percent of female and 61 percent of male respondents had the right to sell their dwelling (either solely or jointly). Rates of other rights, such as the right to sell, show patterns of gender equality. For example, about 61 percent of women claimed that they have the right to rent out the dwelling, while about 56 percent of men made this claim.

Looking at documented ownership, the data show that 2.9 percent of women reported that they have some documentation related to the ownership of their dwelling. The rate is much higher among men: 4.2 percent (Table 2). As shown in Table 3, focusing on respondents who reported they were owners, men are more likely to have any documents than women (about 43 percent versus 32 percent, respectively). The gap is even larger when we focus on formal document ownership (about 20 percent of women compared to 35 percent of men).

4.2. Correlates of housing ownership and rights

To better understand the level of gender equality in housing ownership and rights in low-income areas of Dhaka, we used linear probability model to explore factors that explain men and women's housing ownership status. Specifically, we explored which factors affect both men and women housing ownership in a similar way and which

Table 2
Housing ownership and rights between men and women (%).

Comparison: Ownership and rights M-F	Female	Male	Total
Housing: reported ownership	9.2	9.7	9.5
Housing: documented ownership	2.9	4.2	3.5

Note: tabulation based on all respondents in sample. Weighted average used.

Table 3
Housing ownership and rights between men and women—among reported owners.

Comparison: Ownership and rights M-F	Female	Male	Total
Housing: right to sell	60	61	61
Housing: right to bequeath	57	55	56
Housing: right to use as collateral	53	56	55
Housing: right to rent out	61	56	59
Housing: right to make improvement	62	55	58
Housing: has some ownership document	32	43	37
Housing: has formal document	20	35	28

Note: Tabulation of reported owners. Weighted average used.

ones affect women disproportionately. Focusing on the correlates of reported housing ownership and documented ownership, Table 4 shows the coefficients from linear models: women's reported housing ownership (column 1) and documented ownership (column 2); similar models for men are shown in column 3 and 4. First, we found that women who participate in the labor force are two percentage points less likely to have documented ownership than those who do not participate. Second, perhaps not coincidentally, older women were more likely than their younger counterparts to own housing, both in terms of reported ownership and documented ownership (3 percentage points for an additional 10 years of age for reported ownership and 1 percentage points for documented ownership). Third, higher education of the respondent and their father's education are positively correlated with housing ownership among women (1 percentage point for each additional year of education, respectively, for reported housing ownership). Fourth, women living in slums were 22 percentage points more likely to report owning the house than women living in poor nonslum areas. However, this variable does not affect documented ownership. Along similar lines, women who had lived in the community for more than five years are more likely to report owning housing than those who have not (by 6 percentage points). Fifth, people with larger dwellings were more likely to be reported and documented owners. Sixth, the level of household wellbeing, measured by monthly per capita food consumption, was not significantly correlated with reported ownership of women and barely correlated with documented ownership. Looking at model 3 and model 4, men's ownership shares a few correlates with the women: level of education, living in slums, dwelling size, and length of stay in communities. Thus, correlates that are unique to women's housing ownership models are father's level of education, proportion of male household members, labor force participation (which negatively affect women's likelihood of documented ownership), and household size (which positively affects women's likelihood of documented ownership).

Next, we further examine property rights of women who reported that they were among owners of their dwelling by running linear regressions with key characteristics. In this regard, we explored whether factors that we hypothesize to affect women's housing ownership (as discussed in the research method section) were also correlated with property rights. Table 5 shows the coefficients from a linear model exploring the correlates of women's property rights over their dwellings as: to sell (column 1), to bequeath (column 2), to use as collateral (column 3), to rent out (column 4), and to carry out improvements (column 5).

We found that older women were 8-11 percentage points more likely to have the right to sell their dwelling for every additional 10 years of age, while age of marriage for women is negatively correlated by about 3 percentage points. Women who had lived in a particular location for more than five years were more likely to have the right to sell (by 35 percentage points) than those who have not.

For all outcomes, we found that an additional year in a woman's education is positively associated with an increase in the right to carry out each activity by approximately 1 percentage point. Women's age was also significant across all types of rights. This could underscore the

Table 4
Correlates of ownership status of men and women.

Variables	Female		Male	
	Reported ownership	Documented ownership	Reported ownership	Documented ownership
	(1)	(2)	(3)	(4)
Individual is in the labor force (1/0)	−0.025	−0.022**	0.010	0.014
	(0.016)	(0.010)	(0.045)	(0.029)
Age of respondent	0.003***	0.001**	0.002**	0.001*
	(0.001)	(0.001)	(0.001)	(0.001)
Years of completed education	0.008***	0.003*	0.007***	0.006***
	(0.003)	(0.002)	(0.003)	(0.002)
Married (1/0)	0.006	0.009	0.010	0.022
	(0.029)	(0.013)	(0.101)	(0.066)
Age of marriage	0.003	0.002	0.009***	0.005*
	(0.003)	(0.002)	(0.003)	(0.003)
Household size	0.006	0.010**	0.006	0.000
	(0.006)	(0.005)	(0.006)	(0.004)
Proportion of male members in household	−0.021	−0.056**	−0.017	−0.042
	(0.042)	(0.027)	(0.055)	(0.045)
Father's education (years)	0.011***	0.004*	0.003	0.002
	(0.003)	(0.002)	(0.004)	(0.003)
Mother's education (years)	0.001	0.007	−0.003	−0.006*
	(0.006)	(0.005)	(0.005)	(0.004)
Individual lives in a slum (1/0)	0.219***	0.019	0.227***	0.019
	(0.025)	(0.013)	(0.028)	(0.018)
Total available space in hh (sqft)	0.001***	0.000*	0.001***	0.001***
	(0.000)	(0.000)	(0.000)	(0.000)
Lived in community more than 5 years (1/0)	0.060***	0.017**	0.066***	0.033***
	(0.013)	(0.009)	(0.014)	(0.009)
Monthly per capita food expenditure (ln Taka)	0.011	0.011*	0.004	−0.002
	(0.011)	(0.007)	(0.011)	(0.008)
Constant	−0.410***	−0.184***	−0.523***	−0.239**
	(0.093)	(0.058)	(0.158)	(0.108)
Observations	1,245	1,245	1,025	1,025
R-squared	0.225	0.070	0.226	0.083

Note: Results show coefficients of weighted linear models. Dependent variables are included as column headers. Robust standard errors are in the parenthetical. *, **, and *** indicate significance below the 1 percent, 5 percent, and 10 percent levels.

respect that older dwelling owners command. Similar results were seen for women whose fathers have an additional year of education among the rights to use as collateral, to rent out, and to make improvement. Living in a slum, rather than in a nonslum low-income area, is strongly and positively associated with the right to use as collateral (23 percentage points) and the right to rent out (17 percentage points). Being married was only correlated with the right to rent out the home (36 percentage points). Lastly, women who had lived in an area for more than five years are more likely than those who have not to have rights to sell, to rent out, and to make improvement, the coefficients range around 35–36 percentage points. The findings are consistent with

Table 5
Correlates of rights over dwelling among female reported owners.

Variables	Right to sell the home	Right to bequeath the home	Right to use home as collateral	Right to rent out the home	Right to make home improvements
	(1)	(2)	(3)	(4)	(5)
Individual is in the labor force (1/0)	0.062	0.083	0.014	−0.011	−0.044
	(0.091)	(0.092)	(0.093)	(0.084)	(0.089)
Age of respondent	0.008*	0.009**	0.011**	0.010**	0.008*
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Years of completed education	0.010	0.009	0.018	0.008	0.003
	(0.016)	(0.015)	(0.016)	(0.014)	(0.015)
Married (1/0)	0.138	0.128	0.127	0.355**	0.235
	(0.162)	(0.156)	(0.182)	(0.144)	(0.144)
Age of marriage	−0.029**	−0.030**	−0.020	−0.005	0.000
	(0.014)	(0.014)	(0.015)	(0.015)	(0.015)
Household size	0.014	−0.006	−0.011	−0.034	0.002
	(0.022)	(0.023)	(0.023)	(0.022)	(0.022)
Proportion of male members in household	−0.044	−0.113	−0.092	−0.044	0.195
	(0.271)	(0.272)	(0.243)	(0.257)	(0.254)
Father's education (years)	0.009	0.008	0.012*	0.019**	0.016*
	(0.012)	(0.011)	(0.007)	(0.008)	(0.009)
Mother's education (years)	−0.005	−0.017	−0.036	−0.019	−0.026
	(0.021)	(0.022)	(0.025)	(0.020)	(0.021)
Individual lives in a slum (1/0)	0.130	0.084	0.227**	0.164*	0.106
	(0.097)	(0.096)	(0.091)	(0.095)	(0.097)
Total available space in hh (sqft)	−0.001	0.000	−0.000	−0.001	−0.001
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Lived in community more than 5 years (1/0)	0.354**	0.242	0.231	0.349**	0.362**
	(0.166)	(0.162)	(0.145)	(0.145)	(0.158)
Monthly per capita food expenditure (ln Taka)	0.088	0.049	−0.008	−0.044	−0.035
	(0.066)	(0.065)	(0.065)	(0.061)	(0.062)
Constant	−0.350	−0.055	0.080	0.097	−0.003
	(0.578)	(0.597)	(0.613)	(0.604)	(0.628)
Observations	179	179	179	179	179
R-squared	0.112	0.099	0.145	0.161	0.140

Note: Results show coefficients of weighted linear models. Dependent variables are included as column headers. Sample is restricted to women. Robust standard errors are in the parenthetical. *, **, and *** indicate significance below the 1 percent, 5 percent, and 10 percent levels.

literature describing slum formation in Dhaka: that women who settled in slums earlier are likely to rent out property and access NGO network (which could include micro finance).

Overall, we found that for female housing owners, factors such as higher age, higher self and paternal education, living in a slum rather than in low-income nonslum areas, and tenure of stay in the area (of more than five years) were positively correlated with outcomes such as the right to own and utilize the land as per her needs.

4.3. Reported ownership and risks of eviction

The second hypothesis that we test in this paper is whether housing ownership reduces the perceived risks of eviction and whether the impacts vary across gender. Owners of dwellings in slums and low-income areas, even when they were well established in their neighborhoods and were not worried about appropriation by neighbors, can still face long-term risk of eviction because their dwellings are on public land. Moreover, ownership documents of slum residents tend to be informal in nature and may not grant all the rights that would prevent eviction when government expropriates the land.

This is a critical research question because being evicted is associated with subsequent adverse economic outcomes for low-income households (Desmond 2012) and reduced access to credit and durable consumption (Humphries et al. 2019). Threat of eviction in urban informal settlement reduces investment in housing and infrastructure (Field 2005), further dampened the poor's quality of life. Our study used the data for "concerns about eviction" as reported by respondents as the dependent variable. This variable can be viewed as "perceived" tenure security, as suggested by Barry (2015). The estimation model tests whether reported ownership or documented ownership may affect an individual's perceived housing security.

We conducted this regression analysis using a linear probability model. Table 6 presents coefficients the linear probability models, where the baseline model is shown in column 1. Reported ownership and documented ownership are added in model 2 and model 3, respectively. Subsequently, interaction terms between the ownership variables and female dummy are added to model 4 (reported ownership) and model 5 (documented ownership).

Regression coefficients show that unlike the result from the tabulations, where concern about eviction was much higher among women than men (22 percentage points versus 15 percentage points), after controlling for other characteristics of respondents, the coefficient for sex of respondent was not significant. Other control variables had coefficients with expected signs and significance. Slum residency status was correlated with greater concern about eviction than in nonslum areas by about 28 percentage points, reflecting vulnerability of tenure in slum areas. Higher proportion of male household members was correlated with lesser concern about eviction (1.3 percentage points for 10 percent increment in male members), which could reflect local gender roles in protecting property against eviction. That people who have lived in the community for a long time are less concerned about eviction could be explained by the fact that governments may be less likely to evict communities with established communities with residential infrastructure (Field 2005) and where residents are organized to fight against eviction (Lewis and Hossain, 2017).

The hypothesis that housing ownership reduces the perceived risks of eviction for men and women differently was tested by examining coefficients of the housing ownership variables. The results revealed that reported ownership was not correlated with concerns of eviction (model 2 and model 4), but documented ownership was. In model 3, respondents who have any forms of housing ownership documents were less likely to express concern about eviction (16 percentage points), after controlling for demographic and housing characteristics. Gender differences in the correlation between documented ownership and risk of eviction is explored in model 5. The coefficients show that men with documented ownership in low-income areas of Dhaka were less likely to express concerns about eviction (by 11 percentage points). Moreover, the magnitude of this coefficient for women is more than double that of men; women with documented ownership were about 23 percentage points less likely to express concern about eviction. Such a large gender gap in the security about dwelling implies policies and will be discussed in the next section.

Table 6
Eviction concern regressions.

Variables	Model 1 (1)	Model 2 (2)	Model 3 (3)	Model 4 (4)	Model 5 (5)
Sex of respondent: male = 1; female = 0	−0.036	−0.036	−0.037	−0.037	−0.040
	(0.030)	(0.030)	(0.030)	(0.030)	(0.030)
Individual is in the labor force (1/0)	−0.029	−0.029	−0.032	−0.030	−0.034
	(0.029)	(0.029)	(0.029)	(0.029)	(0.029)
Age of respondent	0.000	0.000	0.000	0.000	0.000
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Years of completed education	−0.002	−0.002	−0.002	−0.002	−0.002
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Married (1/0)	−0.013	−0.013	−0.011	−0.013	−0.011
	(0.049)	(0.049)	(0.049)	(0.049)	(0.049)
Age of marriage	−0.001	−0.001	−0.001	−0.001	−0.001
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Household size	0.004	0.004	0.005	0.004	0.006
	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)
Proportion of male members in household	−0.127**	−0.127**	−0.135**	−0.127**	−0.136**
	(0.060)	(0.060)	(0.060)	(0.060)	(0.060)
Father's education (years)	−0.004	−0.004	−0.004	−0.004	−0.004
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Mother's education (years)	0.007	0.007	0.007	0.007	0.008
	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)
Individual lives in a slum (1/0)	0.278***	0.279***	0.281***	0.279***	0.282***
	(0.025)	(0.025)	(0.024)	(0.025)	(0.024)
Total available space in hh (sqft)	0.000**	0.001**	0.001***	0.001**	0.001**
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Lived in community more than 5 years (1/0)	−0.042*	−0.042*	−0.038	−0.042*	−0.038
	(0.025)	(0.025)	(0.025)	(0.025)	(0.025)
Monthly per capita food expenditure (ln Taka)	−0.017	−0.017	−0.016	−0.017	−0.016
	(0.015)	(0.015)	(0.015)	(0.015)	(0.015)
Reported owner		−0.002		0.006	
		(0.030)		(0.038)	
Documented owner			−0.159***		−0.107***
			(0.032)		(0.039)
Reported owner x female				−0.016	
				(0.054)	
Documented owner x female					−0.118**
					(0.059)
Constant	0.356***	0.355***	0.324***	0.356***	0.327***
	(0.117)	(0.118)	(0.117)	(0.118)	(0.117)
Observations	2,270	2,270	2,270	2,270	2,270
R-squared	0.088	0.088	0.094	0.088	0.094

Note: Results show coefficients of weighted linear models. Dependent variables are included as column headers. Sample is restricted to women. Robust standard errors are in the parenthetical. *, **, and *** indicate significance below the 1 percent, 5 percent, and 10 percent levels

5. Conclusion and Discussion

This research underscores important implications for a rapidly urbanizing country like Bangladesh. Using an innovative method for collecting data on individual level assets (Kilic and Moylan, 2016), the DIGNITY (Dhaka low Income area GeNder, Inclusion and poverTY) presents information on men's and women's housing ownership in low-income areas of Dhaka. The data allows for analysis to better understand gender equality in housing ownership and policy implications.

In terms of data collection, a contribution of this study is the innovation in data collection on a population that is typically overlooked in household surveys: low-income neighborhoods and slum communities. It also presents data about housing, on which there is less information than on other assets. Moreover, our data has information related to women's economic empowerment, such as their participation in decision making and time use in urban setting. The completion of our data collection effort proves that such an undertaking can be accomplished in an urban setting. This innovation could inform future household surveys. In the case of Bangladesh, module about individual asset ownership developed by DIGNITTY could be adopted by the Household Income and Expenditure Survey (HIES) as part of the individual-level interviews when administering the modules on education, health, labor and employment. Such an asset module would allow the country to monitor the SDG indicators 1.4.2 and 5.a.1 (FAO (Food and Agriculture Organization) et al., 2019). By measuring women's asset ownership at the individual level, sex-disaggregated data are available to help stakeholders understand women's empowerment, economic vulnerability in the case of death or divorce, and bargaining power within the household (UN DESA (United Nations Department of Economic and Social Affairs), Population Division, 2019). Individual-level data also allow researchers to learn about married women's property rights (Kieran et al. (2015)). However, stakeholders must recognize that individual-level data is costlier to collect. Interviewing multiple respondents per household often requires several visits to the household and increases the implementation cost of the household survey.

With respect to analytical findings, our study shows the relative gender equality in housing ownership in low-income areas in Dhaka, Bangladesh. The gender difference in reported ownership is also in line with other dimensions of rights, namely rights to sell, to bequeath, to use as collateral, to rent out, and to make improvements. Such a consistency between reported ownership and rights appears to confirm that the relative gender equality in this data is valid. We also examined correlates of women's housing ownership and rights and found that the key correlates are age, education level, father's education level, living in

slums, size of dwelling, and length of stay in the community. Our study also revealed that women are less likely to have documented ownership[7] than men. Even among reported owners, only 32 percent of female owners reported having any types of ownership documents, compared to about 43 percent of male owners. The gap may be explained by constraints to female housing ownership, such as lack of access to government services and gender norms.

There are a few explanations for this finding about relative equality in reported housing ownership in urban areas.[8] First, the direct interview method might have contributed to the higher rate of reported ownership by women. An experiment in rural Uganda found that female respondents are more likely to report joint ownership of dwelling compared to the traditional survey method that ask information from the most knowledgeable household members, often male household heads (Kilic and Moylan, 2016). Second, women in low-income areas of Dhaka often work in paid employment and have means to accumulate wealth. The rate of female labor force participation in low income areas of Dhaka is 58 percent, far higher than the national average of 36 percent. What's more, working women in our sample tend to work paid employment, while many employed women in rural areas work as unpaid family workers (Farole and Cho, 2017; Solotaroff et al., 2019). With monetary income, these women could save and purchase a property. Third, women in low-income areas tend to be aware of their rights and active in community organizations. Combined with greater economic participation, women-centered community-based organizations in the slums have been successful in supporting local improvements and access to services (Lewis and Hossain, 2017). In addition, the model of community organization in slums advocated by the Asian Coalition for Housing Rights (ACHR) often builds on organizing a women's saving group as the first step of community organization. This model has been adopted throughout Asia since 2010's (Buckley and Kallergis, 2014). Finally, the narrow gender gap contrasts well with the gender gap in ownership of agricultural land in rural areas of Bangladesh. Agricultural land is often passed down through inheritance; thus, cultural factors such as patrilineal inheritance encourage giving lands to sons and pressuring women to give up their share to their male relative. Other gender norms further prevent women from claiming agricultural land (Solotaroff et al., 2019). However, such strong gender norms may not be applied in the same extent in low-income areas of Dhaka.

Tenure security and risk of eviction are critical to welfare of poor city dwellers. Eviction could produce negative economic outcomes down the road (Desmond 2012, Humphries et al., 2019), while the threat of eviction reduces housing investment and reduces labor supply (Field, 2005, 2007). Our study tests the research question whether housing ownership reduces the perceived risks of eviction. Since concerns about eviction affected women more than men (22 percent versus 15 percent), we also explore sex-disaggregated effects. We find that, after controlling for other characteristics, there is no difference in eviction concerns between men and women. The regression results also reveal that reported ownership was not correlated with perceived tenure security (as the regression coefficients are insignificant in all models). On the other hand, documented ownership is significantly correlated with perceived tenure security, and the effect was significantly greater among women than men. Among women, having any forms of housing ownership document was correlated with 23 percentage points reduction in eviction concerns, compared to only 11 percentage points among men. This finding underscores the outsized importance of ownership documents for women.

The analytical findings imply crucial policy recommendations. First is the importance of housing ownership documentation. The study shows that even documentation that is not legal but considered "authoritative" in the community can provide some degree of tenure security. The data show that women already have some rights over property and reported that they are owners of property at the same proportion as men. It is important to ensure that they have some form of documents that lends some level of validity from the government. As suggested by UN-Habitat (United Nations Human Settlements Programme), 2004, governments "need to increase rights of occupancy, use, development, etc., for all households in unauthorized settlements, especially for women." Once the situation has been steadier, stakeholders can focus on improving the tenure systems. Authorities may want to explore at least guarantee occupancy rights—equally for both men and women—in informal settlements. In this regard, Nakamura (2017) finds that slum declaration in Pune city of India, although does not give full tenure rights, can increase perception of tenure security and consequently their investment in housing improvement. Similarly, a study of slum regularization in Chandigarh, India shows that documentation also foster women's decision making and empowerment (Datta, 2006). This strategy could also be integrated into a gradual formalization approach, starting with the legal assurance of occupancy rights (Payne, 2001). An example is a model implemented by the Asian Coalition for Housing Rights in communities on government land. The communities negotiate with the government for land on lease or for sale at nominal rates or sometimes even for free. Documentations could also be a title or a long-term user rights provided to the residents for free, with nominal rent, and sell at subsidized, below-market rates, on installments and granted to an individual or the community. In Thailand, some government agencies lease unoccupied land to either community cooperatives or to individual households, usually on 30-year leases and rent of about $3 or $4 per unit per month (ACHR, 2014).

Second, the large gender gaps in documented ownership call for joint registration of both spouses in any housing documentation. Joint titling should be the default option for any regularization interventions, with "mono" titling requiring active opting out, combined with training and sensitization of staff involved. In addition, the concept of joint titling should be expanded to include any documents such as utility or tax bills that can be used as evidence of occupancy rights because the definition of documented ownership in this analysis is essentially the perceived ownership based on such informal documents. In this regard, organizations such as utility companies, tax authorities, or other agencies that issues some forms of documents could be the agents of change in gender equality in housing ownership.

Given the equality in reported ownership, joint titling would benefit women directly since they would enjoy tenure security from documented ownership. The findings that reported ownership are equal between men and women underpin policies and laws to promote gender equality in housing ownership that are less likely to face opposition. But joint titling alone may not be not enough. Weak institutional and individual resistance could make joint titling policy ineffective. For example, in Nicaragua, the joint titles accounted for only 8 percent because of male resistance. Officials may turn a blind eye and allow men to enter names of their brothers or sons in the joint titles (Varley, 2007). Moreover, the lack of mechanisms to ensure that women can hold on to the land could reinforce primary ownership of land by male household heads (Widman, 2014).

In this regard, the joint titling should be accompanied by interventions to address gender bias in administrative procedures and practices. In Latin America, preferential treatment of women in land titling and land reform programs has helped improve women's rights in all spheres (Rahman, 2014). Another experiment from urban Tanzania, where women's documented land ownership is low, shows that a small conditional subsidy can induce households to adopt joint land titles without dampening demand for the title (Ali et al., 2016, O'Sullivan,

[7] The documented ownership definition in this analysis is the perceived ownership based on documents that can be used as evidence of occupancy rights; the rights to rent out, sell and bequeath underline perception of ownership.

[8] The ownership data presented here is the "reported" ownership, which is the situation that the respondent feels they are one of the owners of their dwelling and could be either sole or joint ownership.

2017).

Third, urban upgrading programs must ensure women participation. The data reveal that while women themselves reported that they own housing and hold economic rights, many women do not have ownership documents. Thus, it is important that urban upgrading programs recognize different types of ownership support documents, including informal documents, and housing rights. Not considering informal ownership documents would systematically exclude women, who are de facto one of the owners and rights holders, pushing them out of the conversation in community upgrading. Upgrading programs should assess how any upgrading, infrastructure or legal/regulatory reform project will affect men and women differently. Because impacts can be both positive and negative, affected parties should be consulted and informed about potential costs and benefits before a project is designed and implemented. Participatory methods of recording land rights through community enumeration and street addressing programs can improve security and create a sense of belonging and citizenship (Barry 2015). Moreover, information about ownership and rights should be collected directly from male and female household members instead of only focusing on the head of the households.

Finally, although the DIGNITY survey has been a good effort to collect data about property ownership and rights in urban areas, some crucial data were not collected, such as how owners acquire dwellings, incomes from rentals, and information about landlords. Follow up data collection could help us better understand dynamic of property ownership and women's economic empowerment. This would allow us to further our understanding of the relationship between women's property ownership and other aspects of women's economic empowerment and strengthen the impact of gender equality in housing ownership on welfare and empower the urban poor.

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.landusepol.2020.104983.

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